Analysis of Algorithms
Lab 05

1. Create a program that implements the Insertion Sort Algorithm. Your program has to be able to sort a randomly generated array of alphabetic characters. Decide if you have to do or not any adjustments for the algorithm to work with alphabetic characters instead of integer numbers.
2. Generate a list of 10000000 (ten million) random numbers with C++. Use the Binary Search and the Interpolation Search to find a key. Measure and report the execution time for each search.
3. Prove that the topological sorting problem has a solution if and only if it is a DAG.
4. For a DAG with n vertices, what is the largest number of distinct solutions the topological sorting problem can have? Show your solution.

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.